Sahaj Bhadja

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Summary

Software Engineer with 2+ years in **Python** and **C++**, building system integrations, real-time communication over **TCP/IP** and WebSockets, and automated **validation** pipelines, with hands-on troubleshooting, root cause analysis, and test reporting.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built **data integration** pipelines connecting FastAPI services, SQL Server, PostgreSQL, and cloud storage with incremental loads, **reconciliation checks**, and error handling, sustaining 99%+ reliability across 5K+ monthly production jobs.
- Restructured fault-tolerant asynchronous **Python** job processing with explicit state transitions and failure isolation, tripling throughput across 1K+ jobs and preventing cascading failures under heavy concurrent production load.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built Python pipelines ingesting 100K+ vehicle trajectory logs sampled at 10Hz, aligning ego state, nearby agents, and HD-map signals into a common coordinate frame for **time-series** validation and training.
- Validated a transformer planner through **closed-loop simulation** testing against three baselines, documenting observed failure modes such as off-road trajectories and improving long-horizon planning 18% and stability 15%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Automated nightly **release validation** in Jenkins that built executables, installed them on Windows, Linux, and macOS test VMs, ran 20+ integration and unit suites, and published CSV test reports.
- **Troubleshot** API integrations for **low-bandwidth** users by analyzing failure logs, trimming payloads with pagination and field filtering, and caching hot data, improving response times 30% across distributed clients.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Implemented real-time **WebSocket** communication and RESTful Node.js microservices for a multi-site resource platform, synchronizing 2M+ records across locations with consistent allocation state and fewer cross-site sync conflicts.
- Monitored platform health with **Sentry, Azure Monitor, and Kibana** logs, and gated staging-to-production releases behind automated unit, integration, and end-to-end tests in Jenkins with manual approval steps.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Resolved mobile **connectivity issues** in wallet-signature login and kept booking status synchronized in real time through Firebase across Android, iOS, and web clients, improving cross-platform compatibility 15%.
- Built Flask APIs with **idempotent webhook** handling for payment events, keeping booking responses under 200ms across 50K+ records and cutting manual refund handling about 70% through automated event processing.

Skills

Languages: Python, C++, C, Java, JavaScript, SQL, Bash
Networking & Communication: TCP/IP, REST APIs, WebSockets, Webhooks, JSON, XML, YAML, Real-Time Sync
Testing & Validation: Integration Testing, Unit Testing, PyTest, Mockito, Selenium, Release Validation, Test Automation, Test Reporting
Troubleshooting: Root Cause Analysis, Log Analysis, Debugging, Sentry, Azure Monitor, Kibana, Jenkins, CI/CD
Platforms & Tools: Linux, Windows, macOS, Docker, Git, AWS, Azure, Firebase, Technical Documentation

Projects

Modbus Meter Integration Gateway | *Python, Modbus TCP/RTU, RS-485, MQTT, SNMP, Docker*

- Designed a Python gateway polling **Modbus RTU and Modbus TCP** power meters over RS-485 and Ethernet, normalizing register maps into voltage, current, and kWh readings, and publishing them through MQTT and SNMP.

Device Interoperability Test Harness | *Python, PyTest, Modbus Simulator, Wireshark, TCP/IP*

- Designed a PyTest **interoperability** harness that simulates third-party meters, injects timeouts, bad CRCs, and address conflicts, captures TCP/IP traffic, and generates validation reports for **firmware release** testing.

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, EJ-2, OS Scheduling*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel queue scheduling in **C++** with React controls and Gantt visualization, testing 1K+ process combinations and cutting average turnaround time 25%.